Supplementary Material: Deep Architectures Can Generalize: Sparse Feature Selection Improves Spatial Transfer in Earth Observation

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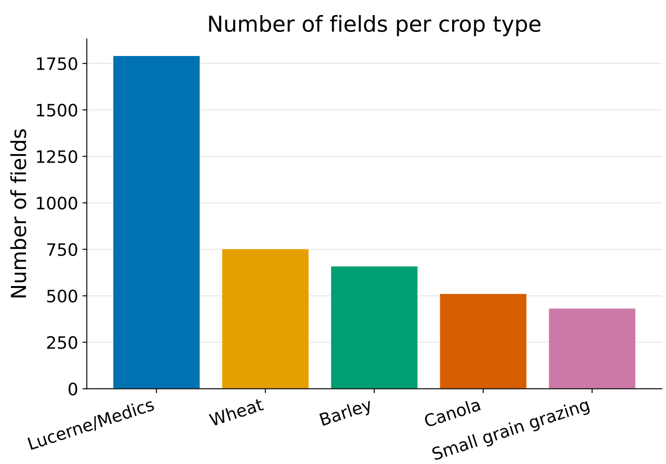

Fig. S1. Number of fields per crop type in the training and holdout regions.

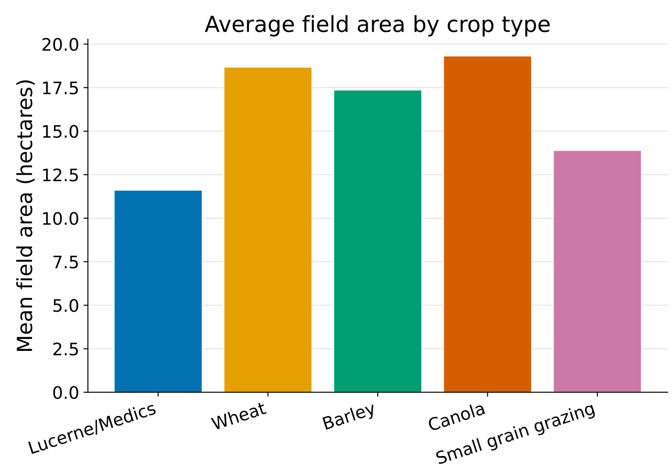

Fig. S2. Average field area by crop type (hectares).

This document collects supporting material for the main paper. Section S-A details the dataset, feature set, and model definitions; Section S-B defines the metrics; Section S-C reports training cost; Section S-D presents the ablation studies; Section S-E the data-efficiency experiment; and Section S-F the raw-versus-`xr_fresh` representation comparison. Equation, figure, table, and section labels are prefixed "S".

S-A. Dataset Details, Feature Set, and Model Definitions

Supports main paper Sec. II–III.

A. Exploratory Data Analysis

The five crop classes are heavily imbalanced (Fig. S1). Lucerne/medics dominates both regions despite having the smallest mean field area (Fig. S2). Spectral responses overlap substantially across crops (Fig. S3), and wheat and barley are especially similar (Fig. S4), which is the dominant source of persistent confusion noted in the main paper.

B. `xr_fresh` Feature Set

Table S1 lists the time-series features extracted from each pixel's 10-month spectral profile with `xr_fresh` [1], [2], forming the 174-dimensional per-pixel vector used by the classical learners (Fig. S5 illustrates the extraction).

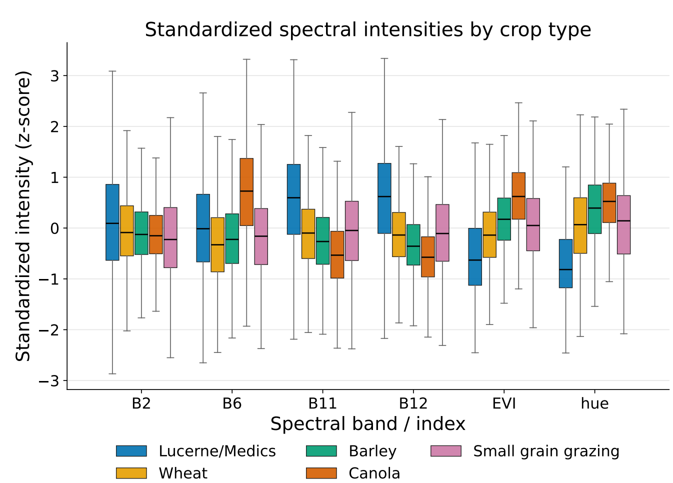

Fig. S3. Spectral band intensities by crop type. The substantial overlap shows that classification from a limited set of spectral bands alone is challenging.

C. Baseline Classical Model Definitions

For a feature vector $\mathbf{x}_i$ and class $c \in \{1, \ldots, C\}$, the four baselines are as follows. Logistic regression [3] (multinomial):

$$P(y_i = c \mid \mathbf{x}_i) = \frac{\exp(\mathbf{w}_c^\top \mathbf{x}_i + b_c)}{\sum_{c'=1}^{C} \exp(\mathbf{w}_{c'}^\top \mathbf{x}_i + b_{c'})}. \tag{S1}$$

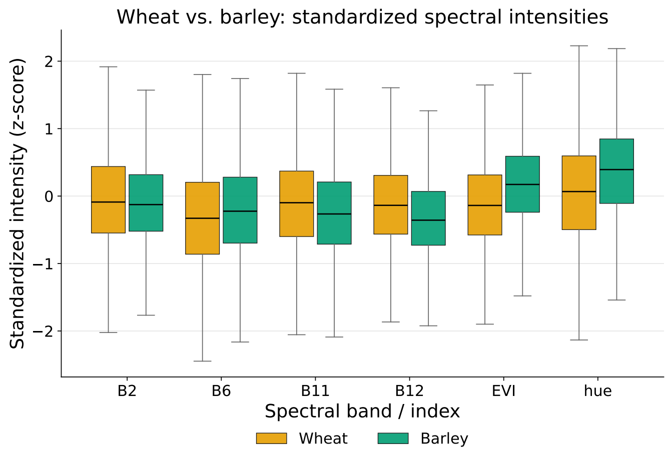

Fig. S4. Wheat versus barley spectral band intensities. The high similarity motivates their persistent classification confusion across all models.

TABLE S1
TIME-SERIES FEATURES EXTRACTED FROM EACH PIXEL'S SPECTRAL PROFILE WITH XR_FRESH.

Feature	**Description**
Absolute Energy	Sum of squared values.
Absolute Sum of Changes	Total absolute difference between consecutive measurements.
Autocorrelation (1, 2-month lag)	Correlation with one- and two-month lagged versions.
Count Above Mean	Number of points above the series mean.
Day of Year of Max/Min	Julian day of the maximum and minimum.
Kurtosis	Tail heaviness of the distribution.
Linear Trend	Slope of a least-squares fit.
Longest Strike Above/Below Mean	Longest consecutive run above or below the mean.
Max/Min	Absolute extreme values.
Mean/Median	Measures of central tendency.
Mean Absolute Change/Mean Change	Average magnitude and signed change between points.
Mean Second Derivative	Mean of the series' second differences.
Quantiles (0.05, 0.95)	5th and 95th percentiles.
Ratio Beyond $r\sigma$	Proportion of points more than r s.d. ($r = 1, 2, 3$) from the mean.
Skewness	Asymmetry around the mean.
Standard Deviation/Variance	Spread measures.
Sum of Values	Total sum of observations.
Symmetry Looking	Similarity under series reversal.
Time Series Complexity	Complexity-Invariant Distance metric of irregularity.

Random Forest [4]: majority vote over K trees, $\hat{y}_i = \mathrm{mode}\{T_1(\mathbf{x}_i), \ldots, T_K(\mathbf{x}_i)\}$. LightGBM [5]: additive boosting with rate η, $\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta\, f_t(\mathbf{x}_i)$. XGBoost [6], [7]: regularized boosting objective $\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$ with $\Omega(f_k) = \gamma T_k + \frac{1}{2}\lambda \sum_j w_j^2$. The field-level XGBoost was tuned with Optuna over 150 trials; the top configurations appear in Table S2. Field-level pixel features are aggregated by the mean within each polygon (Fig. S6). For the pixel ensemble, RF, XGBoost, and LightGBM probabilities are soft-voted; the field-level stacked ensemble uses RF and XGBoost base learners with a logistic-regression meta-learner, and a confidence-margin rule falls back to pixel-mode voting when the top two field-level class probabilities differ by less than 0.10.

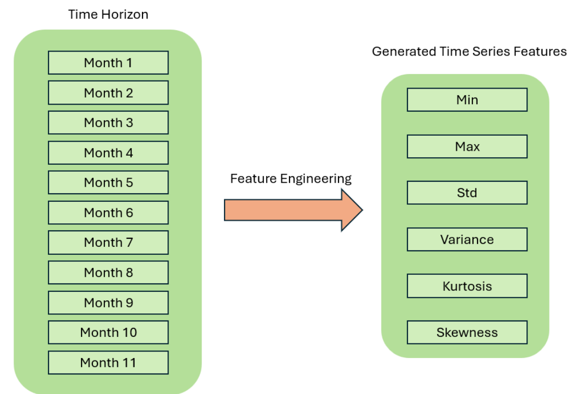

Fig. S5. Extraction of statistical and temporal features from the 10-month spectral time series.

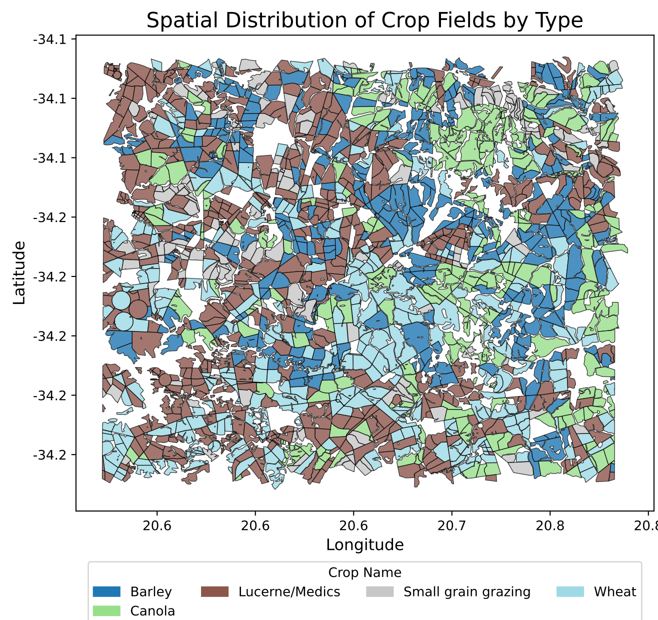

Fig. S6. Field-level aggregation of pixel features: pixel values are grouped by field polygon and summarized into a single feature vector per field.

D. Deep Model Definitions

a) CNN-BiLSTM: An input tensor of shape $(B, 6, T)$ passes a 1D convolution (64 filters, kernel 5) and a bidirectional LSTM (hidden size 64), with the final timestep classified via a softmax head (Fig. S7). Class imbalance is handled with focal loss $\mathcal{L}_{\text{focal}} = -\alpha_t(1-p_t)^\gamma \log(p_t)$, $\gamma = 2.0$, α_t the inverse class frequency. Five seeds are ensembled by averaging logits.

b) TabNet: At each of n_{steps} decision steps, an attentive transformer produces a sparse feature mask via *sparsemax*,

$$\begin{aligned} \mathbf{M}[i] &= \mathrm{sparsemax}\big(\mathbf{P}[i-1] \odot h_i(\mathbf{a}[i-1])\big), \\ \mathbf{P}[i] &= \textstyle\prod_{j=1}^{i} \big(\gamma - \mathbf{M}[j]\big), \end{aligned} \tag{S2}$$

TABLE S2
XGBOOST HYPERPARAMETER SEARCH: TOP 5 OPTUNA TRIALS WITH VALIDATION ACCURACY.

Trial	Acc.	Est.	Depth	LR	SubS	ByTree	Gamma	Regα	Regλ	MinCW
32	**76.61%**	1141	13	0.0215	0.5749	0.6986	0.5589	0.00184	0.00259	5
12	76.46%	984	12	0.0145	0.6675	0.6946	0.0799	0.00503	0.00455	7
24	76.40%	1165	11	0.0162	0.7831	0.6403	0.00413	0.00481	0.00478	9
13	76.29%	995	12	0.0277	0.6345	0.6829	0.3999	0.00011	0.00567	6
18	76.29%	915	13	0.0121	0.6784	0.5958	0.1612	0.00491	0.1732	8

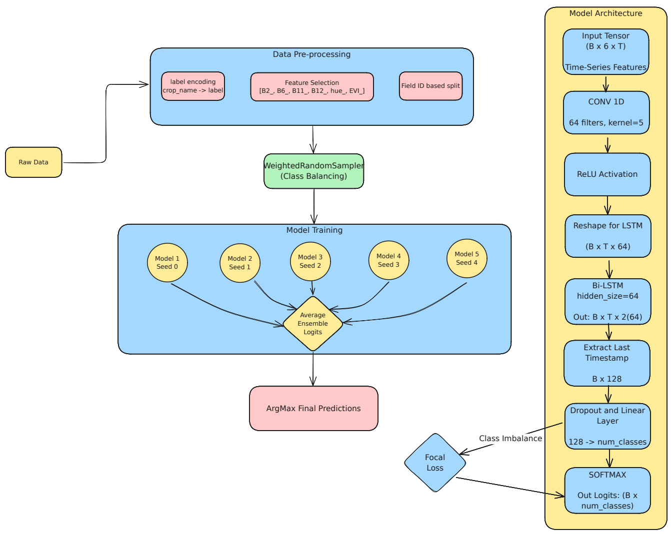

Fig. S7. Architecture of the CNN + BiLSTM model: 1D convolutional temporal feature extraction followed by a bidirectional LSTM.

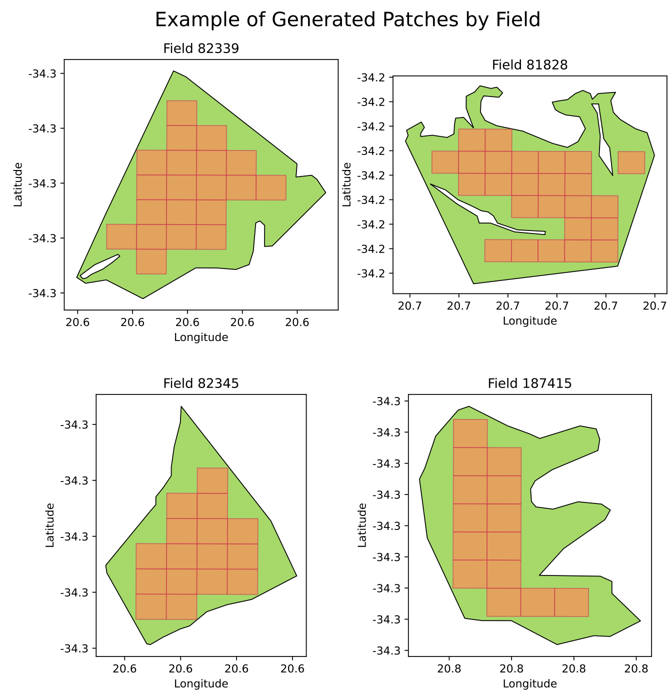

Fig. S8. Illustration of patch generation per field.

with relaxation $\gamma = 1.5$; the masked features are handled by a feature transformer. We use $n_d = n_a = 64$, 5 decision steps, batch size 1024, and ensemble five seeds by averaging softmax outputs. The sparsemax masks are the mechanism the main paper's L-TAE-S grafts onto a temporal-attention backbone [8], [9].

c) L-TAE: Given an embedded sequence $\{\mathbf{z}_t\}_{t=1}^{T}$, a learned master query attends over time; for a single head

$$\mathbf{o} = \sum_{t=1}^{T} \alpha_t \, \mathbf{v}_t, \qquad \alpha_t = \frac{\exp(\mathbf{q}^\top \mathbf{k}_t / \sqrt{d})}{\sum_{t'} \exp(\mathbf{q}^\top \mathbf{k}_{t'} / \sqrt{d})}, \tag{S3}$$

with channel grouping keeping the encoder lightweight [10]. Evaluated at pixel and field level as a five-seed ensemble.

d) Transformer encoder: After the same band embedding and month-aware positional encoding as L-TAE, the sequence passes three pre-norm multi-head self-attention layers ($d_{\text{model}} = 128$, 8 heads, feed-forward width 256), is mean-pooled, and classified [11], [12]. Unlike L-TAE-S it has no feature-sparsity mechanism, serving as the capacity-versus-sparsity control. Five-seed ensemble under the same weighted-focal objective.

e) TempCNN: Stacked 1D convolutions along the temporal axis with batch normalization and dropout, followed by a softmax head [13]; pixel and field variants, five seeds.

f) Patch models: Each field is tiled into 100 m patches ($\sim$10 $\times$ 10 pixels) resampled to 128 $\times$ 128 (Fig. S8). The multi-channel 2D CNN treats each band$\times$month as a channel ($C = 60$) through two convolutional blocks; the 3D CNN treats each patch as a spatiotemporal volume ($T = 10$, $C = 6$) through three 3D-convolutional blocks with a weighted focal loss (Fig. S9). Both are five-seed ensembles aggregated to a field label by majority vote.

S-B. PERFORMANCE METRIC DEFINITIONS

Supports main paper Sec. III. Let N be the number of samples, C the classes, and n_c the support of class c. Cohen's Kappa is $\kappa = (p_o - p_e)/(1 - p_e)$ with p_o the observed agreement and $p_e = \sum_c (n_{c,\text{true}}/N)(n_{c,\text{pred}}/N)$. Macro-F1 (primary) and weighted F1 are

$$\text{F1}_m = \frac{1}{C} \sum_{c=1}^{C} \text{F1}_c, \quad \text{F1}_\text{w} = \sum_{c=1}^{C} \frac{n_c}{N} \text{F1}_c, \tag{S4}$$

with $\text{F1}_c = 2\,\text{Prec}_c\,\text{Rec}_c/(\text{Prec}_c + \text{Rec}_c)$. Cross-entropy (Xent) is the AI4FoodSecurity field-level scoring metric [14]; because submissions are hard labels, it is evaluated on the one-hot predicted label clipped to $[\epsilon, 1-\epsilon]$, $\epsilon = 10^{-7}$:

$$\begin{gathered} \text{Xent} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} y_{i,c} \, \ln \hat{q}_{i,c}, \\ \hat{q}_{i,c} = \text{clip}\big(\mathbb{1}[\hat{y}_i = c],\, \epsilon,\, 1-\epsilon\big). \end{gathered} \tag{S5}$$

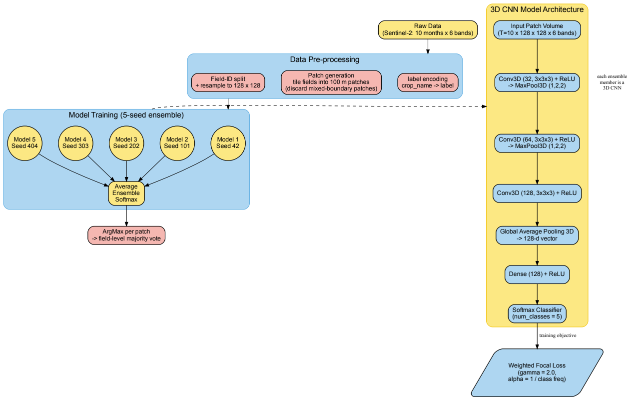

Fig. S9. Patch-level 3D CNN pipeline: each field is tiled into 100 m patches resampled to 128×128; a five-seed ensemble averages softmax outputs, and per-patch predictions are aggregated to a field label by majority vote.

S-C. Training Cost

Supports main paper Sec. III and the Discussion. Table S3 reports wall-clock training time for a single run of each model on one workstation (RTX 3080 Ti). The robust models that transfer best (the `xr_fresh` tree ensembles and the field-level L-TAE-S) are among the cheapest to train, while the most expensive (the patch CNNs and the pixel-level Transformer) do not lead the holdout ranking.

S-D. Ablation Studies

Supports main paper Sec. V ("Two Robustness Levers"). We attribute the holdout ranking primarily to two levers—built-in sparse feature selection and field-level aggregation—and show that ensemble size and the imbalance-handling objective are second-order: they shift absolute scores but never reorder the robust and fragile model groups.

(i) Field-level aggregation (main Table II) recovers most of the transfer gap for dense temporal nets (L-TAE $-0.20 \rightarrow -0.07$; L-TAE-S $-0.17 \rightarrow -0.04$) but harms TabNet ($0.60 \rightarrow 0.43$). *(ii) Sparse selection versus attention capacity*: a plain Transformer with far greater capacity transfers no better than L-TAE (both ST $\mathrm{F1}_m = 0.58$), whereas grafting a sparse channel gate (L-TAE-S) lifts transfer to 0.60. *(iii) Resampling*: the *Stacking* and *SMOTE Stacked* ensembles differ only in SMOTETomek resampling; resampling widens the gap ($-0.11 \rightarrow -0.21$) and lowers holdout $\mathrm{F1}_m$ ($0.53 \rightarrow 0.49$).

(iv) Ensemble size (Table S4): seed-ensembling gives a small, monotone lift but does not change the ordering—a single-seed L-TAE-S ($\mathrm{F1}_m = 0.56$) already exceeds the five-seed CNN-BiLSTM and TabNet field ensembles. *(v) Imbalance-handling objective* (Table S5): the robust L-TAE-S is almost invariant across five loss/sampler settings (0.59–0.60), whereas the fragile CNN-BiLSTM is sensitive (0.41–0.56); even CNN-BiLSTM's best setting (0.56) stays below L-TAE-S's worst (0.59), so the objective never reorders the two.

TABLE S3
TRAINING TIME FOR A SINGLE RUN (ONE SEED) OF EACH MODEL. TREES TIMED AS A SINGLE BEST-CONFIGURATION FIT, EXCLUDING THE ONE-TIME OPTUNA SEARCH. †MEAN PER SEED FOR A 5-SEED ENSEMBLE. ‡MEASURED ON THE 75% TRAINING FRACTION, THE BEST-TRANSFERRING TABNET PIXEL CONFIGURATION.

Model	**Level**	**Features**	**Time / run**
Classical, field-level			
XGBoost	field	`xr_fresh`	12 s
LightGBM	field	`xr_fresh`	31 s
Voting	field	`xr_fresh`	109 s
Stacking	field	`xr_fresh`	109 s
SMOTE Stacked	field	`xr_fresh`	99 s
Classical, pixel-level			
XGBoost	pixel	`xr_fresh`	31 s
LightGBM	pixel	`xr_fresh`	31 s
Logistic Regression	pixel	`xr_fresh`	8.0 min
Random Forest	pixel	`xr_fresh`	73 min
Deep, field-level (raw sequences)			
TabNet (`xr_fresh`)	field	`xr_fresh`	34 s
TabNet (raw, field-avg)	field	raw	39 s
CNN-BiLSTM	field	raw	53 s
L-TAE	field	raw	61 s
TempCNN	field	raw	64 s
L-TAE-S	field	raw	18 s†
Deep, pixel-level (raw sequences)			
CNN-BiLSTM	pixel	raw	10 min†
L-TAE-S	pixel	raw	24 min†
TempCNN	pixel	raw	29 min
L-TAE	pixel	raw	38 min
TabNet (raw)	pixel	raw	61 min†‡
Transformer	pixel	raw	97 min
Deep, patch-level (raw sequences)			
3D CNN	patch	raw	22 min†
Multi-Channel 2D CNN	patch	raw	38 min†

TABLE S4
ABLATION (IV): SPATIAL-TRANSFER HOLDOUT PERFORMANCE VERSUS SEED-ENSEMBLE SIZE, BY RE-INFERENCE OF SAVED PER-SEED CHECKPOINTS. SINGLE-SEED ROWS REPORT MEAN±S.D. ACROSS FIVE SEEDS.

Model	**Ensemble**	$\mathbf{F1}_m$	κ
TabNet (field)	1 seed	0.31 ± 0.02	0.18 ± 0.03
	3 seeds	0.39	0.29
	5 seeds	0.43	0.34
L-TAE-S (field)	1 seed	0.56 ± 0.02	0.47 ± 0.02
	3 seeds	0.58	0.50
	5 seeds	0.60	0.52
CNN-BiLSTM (field)	1 seed	0.48 ± 0.02	0.31 ± 0.02
	3 seeds	0.52	0.34
	5 seeds	0.51	0.33

S-E. Data Efficiency Under Spatial Transfer

Supports main paper contribution 5 (data efficiency), reported in Sec. V. Retraining a representative set on random subsets of 75%, 50%, and 25% of the training fields and re-evaluating on the holdout (Fig. S10) shows the most robust models are also the most data-efficient: TabNet retains essentially full holdout accuracy down to a quarter of the fields ($\mathrm{F1}_m = 0.60, 0.62, 0.62, 0.59$ at $100/75/50/25\%$), logistic regression is similarly flat (0.56–0.60), and L-TAE-S tracks them ($0.60, 0.61, 0.56, 0.58$), whereas the dense pixel L-TAE is noisier and stays below the best transferers throughout

TABLE S5
ABLATION (V): EFFECT OF THE IMBALANCE-HANDLING OBJECTIVE AND RESAMPLING ON SPATIAL-TRANSFER HOLDOUT PERFORMANCE (FIELD-LEVEL, 5-SEED ENSEMBLES). "RESAMPLE" DENOTES CLASS-BALANCED OVERSAMPLING.

Model	Loss	Resample	F1$_m$	κ
TabNet (field)	Weighted focal (γ=2)	no	0.43	0.34
	Weighted CE (γ=0)	no	0.42	0.33
	Plain CE	no	0.35	0.25
	Weighted focal (γ=2)	yes	0.40	0.23
	Plain CE	yes	0.46	0.37
L-TAE-S (field)	Weighted focal (γ=2)	no	0.60	0.52
	Weighted CE (γ=0)	no	0.60	0.50
	Plain CE	no	0.59	0.50
	Weighted focal (γ=2)	yes	0.60	0.49
	Plain CE	yes	0.60	0.50
CNN-BiLSTM (field)	Weighted focal (γ=2)	no	0.51	0.34
	Weighted CE (γ=0)	no	0.49	0.34
	Plain CE	no	0.41	0.30
	Weighted focal (γ=2)	yes	0.56	0.39
	Plain CE	yes	0.50	0.34

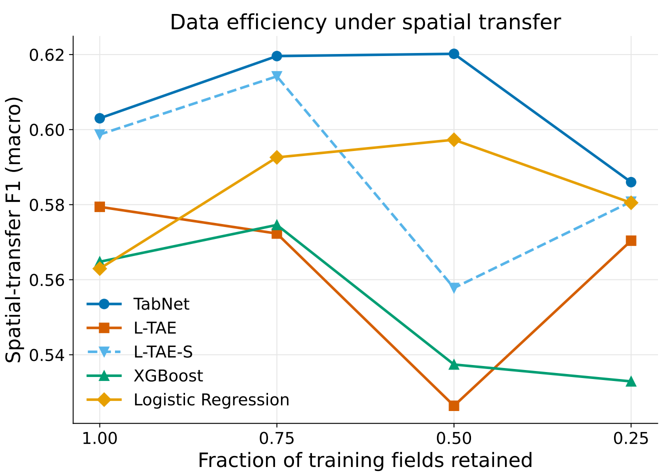

Fig. S10. Spatial-transfer macro-F1 versus fraction of training fields. Robust models (TabNet, logistic regression, L-TAE-S) are nearly flat.

$(0.58, 0.57, 0.53, 0.57)$. Holdout accuracy is therefore limited far more by spatial transfer than by training-set size for the robust models.

S-F. RAW REFLECTANCE VERSUS `xr_fresh`

Supports the raw-versus-`xr_fresh`* supporting experiment discussed in Sec. V.* Because the best-transferring classical learners in main Table I are exactly those fed `xr_fresh` statistics, representation and model family are confounded there. To separate them we trained a single XGBoost learner once on raw monthly reflectance and once on per-pixel `xr_fresh` statistics, holding the FID-wise splits, preprocessing, and field-level mean pooling fixed (Table S6). The two representations are essentially indistinguishable under transfer (F1$_m$ = 0.59 raw vs. 0.56 `xr_fresh`; equal on weighted-F1 and κ; `xr_fresh` marginally better on cross-entropy). The `xr_fresh` representation therefore yields no measurable transfer-accuracy gain over raw reflectance for the trees; its value is operational—a compact, fast-to-fit, agronomically interpretable vector (Sec. S-C)—rather than an accuracy advantage. Figure S11 visualizes the pixel-versus-field aggregation effect summarized in main Table II.

TABLE S6
RAW REFLECTANCE VERSUS `XR_FRESH` FOR A SINGLE FIXED XGBOOST LEARNER, IDENTICAL SPLITS AND POOLING; ONLY THE INPUT REPRESENTATION DIFFERS.

Representation	Split	F1$_m$	wF1	κ	Xent
Raw reflectance	In-region	0.73	0.80	0.71	0.65
`xr_fresh`	In-region	0.72	0.76	0.68	0.74
Raw reflectance	Spatial transfer	0.59	0.69	0.50	0.85
`xr_fresh`	Spatial transfer	0.56	0.69	0.50	0.82

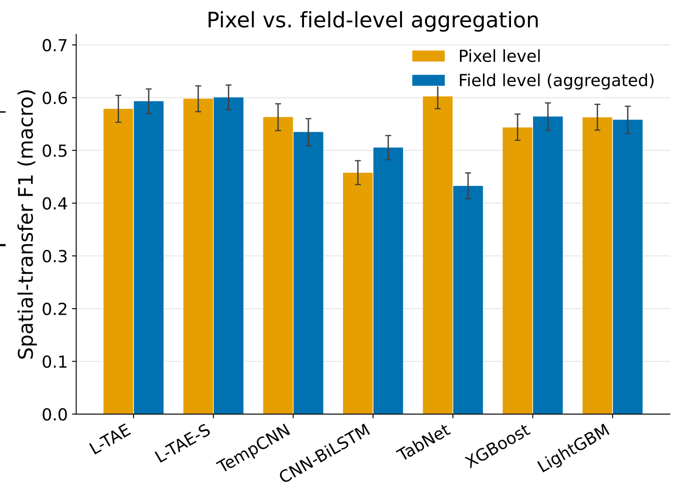

Fig. S11. Spatial-transfer macro-F1 at pixel versus field level. Field-level aggregation recovers transfer for dense temporal models (L-TAE, CNN-BiLSTM) but collapses the sparsity-reliant TabNet. Error bars are 95% bootstrap CIs over holdout fields.

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